

Recall An $r \times c$ matrix A defines a map $f_A: \mathbb{R}^c \rightarrow \mathbb{R}^r$
by formula $f_A(x) = Ax$

$\text{Ker}(f_A) := \{x \mid Ax = 0\}$ is a subspace of $\mathbb{R}^c$

$\text{Im}(f_A) := \{Ax \mid x \in \mathbb{R}^c\}$ is a subspace of $\mathbb{R}^r$

We saw $\dim(\text{Ker } f_A) = \# \text{ of nonpivot cols in RREF}(A)$

$\dim(\text{Im } f_A) = \# \text{ of pivot cols in RREF}(A) =: \text{rank}(f_A)$

Thm $\dim(\text{Ker } f_A) + \dim(\text{Im } f_A) = \dim(\text{domain of } f_A)$

Want to prove: let V, W be finite diml vector spaces and
let $T: V \rightarrow W$ be a "linear" map. (defn below)

Then $\dim(\text{Ker } T) + \dim(\text{Im } T) = \dim V$

What is a linear map? Observe the following properties for f_A

$$\textcircled{1} f_A(v_1 + v_2) = A(v_1 + v_2) = Av_1 + Av_2 = f_A(v_1) + f_A(v_2)$$

$$\textcircled{2} f_A(cv_1) = A(cv_1) = cAv_1 = cf_A(v_1)$$

Defn A map $T: \text{v.space } V \rightarrow \text{v.space } W$ is called
linear if it satisfies

$$\left. \begin{array}{l} \textcircled{1} T(v_1 + v_2) = T(v_1) + T(v_2) \\ \textcircled{2} T(cv_1) = cT(v_1) \end{array} \right\} \begin{array}{l} \text{for all } v_1, v_2 \in V \\ \text{and } c \in \mathbb{R} \end{array}$$

Rmk (I) Thus a linear map T is a map that preserves the
vector space structure i.e. T respects vector addition and
scalar multiplication: "adding first in V and then applying
 T is the same as first applying T to both vectors and
then adding in W ." Similarly for scalar multiplication.

$$(I) \text{ For a linear map } T, \quad T(c_1 v_1 + c_2 v_2) = T(c_1 v_1) + T(c_2 v_2) \\ = c_1 T(v_1) + c_2 T(v_2)$$

$$\text{Similarly } T\left(\sum_{i=1}^n c_i v_i\right) = \sum_{i=1}^n c_i T(v_i)$$

Thus a linear map T "preserves linear combinations".

$$(II) \text{ For linear } T: V \rightarrow W \quad (i) \quad T(\vec{0}) = T(0 \cdot \vec{0}) = 0 T(\vec{0}) = \vec{0}$$

$$(ii) \ker T := \{v \in V \mid Tv = 0\} \text{ is a subspace of } V$$

$$\left[\begin{array}{l} \text{Pf. } v_1, v_2 \in \ker T \Rightarrow T(c_1 v_1 + c_2 v_2) = c_1 T(v_1) + c_2 T(v_2) = 0 + 0 = 0 \\ \text{and } 0 \in \ker T \end{array} \right]$$

$$(iii) \operatorname{Im} T = \{T(v) \mid v \in V\} \text{ is a subspace of } W$$

$$\left[\begin{array}{l} \text{Pf. Let } w_1, w_2 \in \operatorname{Im} T \text{ i.e. } w_1 = T(v_1), w_2 = T(v_2) \text{ for some } v_i \in V \\ \text{We need to show } c_1 w_1 + c_2 w_2 \in \operatorname{Im}(T) \end{array} \right]$$

$$c_1 w_1 + c_2 w_2 = c_1 T(v_1) + c_2 T(v_2) = T(c_1 v_1) + T(c_2 v_2) = T(c_1 v_1 + c_2 v_2)$$

$$\text{Thus } c_1 w_1 + c_2 w_2 \in \operatorname{Im}(T) \quad]$$

$$(IV) \text{ If } S \subset V \text{ spans } V, \text{ then } T(S) \text{ spans } T(V) \text{ i.e. } \operatorname{Im} T$$

Pf

$\Downarrow$

$$\text{given } v \in V, \quad v = \sum_i c_i u_i \text{ for } u_i \in S$$

$\Downarrow$

$$T(v) = \sum c_i T(u_i)$$

$$(V) \text{ If } S \subset V \text{ is linearly } \text{dependent} \text{ then } T(S) \text{ is } \text{dependent.}$$

Contrapositive:

For $S \subset V$, if $T(S)$ is independent then S is also independent

Examples of linear maps

0) For $r \times c$ matrix A ,

$$f_A: \mathbb{R}^c \rightarrow \mathbb{R}^r \text{ is linear} \\ x \mapsto Ax$$

Exercise Show that every linear map $\mathbb{R}^c \xrightarrow{T} \mathbb{R}^r$ is of form f_A
i.e. it is given by $T(x) = Ax$ for a (unique) $r \times c$ matrix. $\square$

1) $V =$ set of all differentiable functions $\mathbb{R} \rightarrow \mathbb{R}$
 $W =$ set of all functions $\mathbb{R} \rightarrow \mathbb{R}$

$D: V \rightarrow W$ is a linear map b/c $(f+g)' = f' + g'$
 $f \mapsto f' =$ derivative of f $(cf)' = cf'$

$\ker D =$ Constant functions

1)' Consider the same map on the subspace of polynomials.

$P :=$ polynomials $\subset V$ $D(P) = P$

$P_n =$ polynomials of $\deg \leq n$ $D(P_n) = P_{n-1}$

$P_n \xrightarrow{D} P_{n-1}$	$\begin{matrix} \dim \downarrow \\ n+1 \end{matrix}$	$\downarrow$ n
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$\dim \ker = 1$, $\dim P_n = n+1$

See that the dimension formula holds.

2) $V =$ set of $n \times n$ matrices $T: V \rightarrow \mathbb{R} = W$
 $W = \mathbb{R}$ $A \mapsto \text{tr}(A)$

Exercise Check that the dimension formula holds.

Find a basis for the kernel.

Added after class: we will prove dimension formula in next class.

Exercises • Can you give more examples of linear maps?

• Easy and important $\left\{ \begin{array}{l} \text{If } V \xrightarrow{f} W \xrightarrow{g} U, f \text{ and } g \text{ linear} \\ \text{then } V \xrightarrow{g \circ f} U \text{ is also linear} \end{array} \right.$